

MCU / SPI control input

3V3 distribution, pullups, and 4-wire mode

R4
3V3
0R
EPD_3V3
Fit: distributes J3 external EPD 3V3 input to EPD_3V3

R6
EPD_3V3
0R
VDD

R8
3V3
0R
V_B

R7
EPD_3V3
0R
VDDIO

R12
GND
0R
BS0
populate: BS0 low, 4-wire

R13
GND
0R
BS1
populate: BS1 low, 4-wire

C23
V_B
10uF/25V/1210
GND

C3
EPD_3V3
0.1uF/50V/0603
GND

R9
TSCL
4.7k 1%
3V3

R10
TSDA
4.7k 1%
3V3

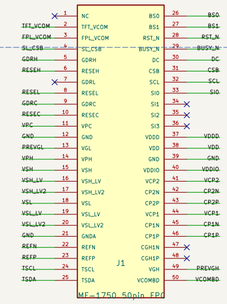
R15
CSB
DNP
SL_CSB
DNP: independent CS configuration verified on real panel

R16
CS2
0R
SL_CSB
Fit: routes GPIO10/CS2 to SL_CSB; real-panel verified configuration

J3
3V3
EPD_3V3_IN
GND
EPD_3V3_IN

R12/R13 set BS0=0 and BS1=0 for 4-wire SPI.

Panel FPC and MF-1750 signals



HV boost and negative gate pump

C0
V_B
4.7uF/25V/0603
GND

L1
V_B
15uH/3.1A SMMS0650
SW_H

D1
VPH
MBR0530
pad1/K=VPH; pad2/A=SW_H
SW_H

C2
VPH
10uF/25V/1210
GND

R5
VPH
0R
PREVGH

C4
SW_H
10uF/25V/1210
NEG_PUMP_H

D2
GND
MBR0530
pad1/K=GND; pad2/A=NEG_PUMP_H
NEG_PUMP_H

D3
NEG_PUMP_H
MBR0530
pad1/K=NEG_PUMP_H; pad2/A=PREVGL
PREVGL

C5
PREVGL
10uF/25V/1210
GND

Q1
GDRH
SI1308EDL
SW_H
RESEH
Vishay SI1308EDL SC-70 top view: pin1=G, pin2=S, pin3=D

R1
RESEH
0.2R 0.2W 1%
GND

Main current path: V_B -> L1 -> SW_H/Q1/D1 -> VPH -> R5 -> PREVGH.

Panel-side caps, references, and flying capacitors

C13
TFT_VCOM
4.7uF/10V/0805
GND

C14
FPL_VCOM
0.47uF/50V/1210
GND

R2
RESEL
0R
GND

C7
VSH
10uF/25V/1210
GND

C8
VSH_LV
10uF/25V/1210
GND

C9
VSH_LV2
10uF/25V/1210
GND

C10
VSL
10uF/25V/1210
GND

C11
VSL_LV
10uF/25V/1210
GND

C12
VSL_LV2
10uF/25V/1210
GND

C21
REFN
1uF/50V/0805
GND

C20
REFP
1uF/50V/0805
GND

C1
VDDD
2.2uF/25V/0603
GND

C19
VCP2
1uF/50V/0805
GND

C18
CP2N
1uF/50V/0805
CP2P

C17
VCP1
1uF/50V/0805
GND

C16
CP1N
1uF/50V/0805
CP1P

C15
VCOMBD
0.47uF/50V/1210
GND

VPC positive pump

C22
V_B
4.7uF/25V/0603
GND

Q2
GDRC
AO3401A
SW_C
V_B
SOT-23: pin1=G, pin2=S, pin3=D

L2
SW_C
15uH/3.1A SMMS0650
RESEC

D4
SW_C
MBR0530
pad1/K=SW_C; pad2/A=VPC
VPC

C6
VPC
10uF/25V/1210
GND

R3
RESEC
0.22R 0.2W 1%
GND

Q2/L2/D4/R3 form the VPC pump; keep this loop compact in PCB layout.

Bring-up and reserved test pads

TP1
3V3
3V3

TP2
GND
GND

TP3
VPH
VPH

TP4
PREVGH
PREVGL

TP5
PREVGL
PREVGL

TP6
VPC
VPC

TP7
VDDIO
VDDIO

TP8
CSB
CSB

TP9
SL_CSB
SL_CSB

TP10
BUSY_N
BUSY_N

Reference source: MF-1750/M009F-T specification page 10 reference circuit.

Verified topology: R16 fitted routes SL_CSB to DevKitC GPIO10; R15 is DNP.

Functional blocks now group and wire local nodes; net labels are reserved for cross-block interfaces.

Default bus mode: BS0=0 and BS1=0 through R12/R13 for 4-wire SPI with D/C.

Diode polarity is assigned in BOM notes; match footprint pin A/K to those notes before PCB layout.

Reserved FPC pins are routed only to isolated single test pads; pin1 NC is explicitly no-connect.

62 x 58 mm engineering layout; R16 independent CS

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File: mf1750_devkitc_b1_single_power_layer.kicad_sch

Title: MF-1750 DevKitC Compact Carrier Rev B1 4L

Size: A2 | Date: | Rev:

KiCad E.D.A. 10.0.3 | Id: 1/1